## The Multiplication Rule

Name _____

You guess on 3 multiple-choice questions. Each has 4 choices. What is the probability you get all 3 right? Each guess has a $\frac{1}{4}$ chance. The questions do not affect each other, so multiply:

$$P(\text{all 3 right}) = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{64}$$

**Independent events.** Two events are **independent** if one happening does not change the probability of the other.

| **Independent:** $P(A \cap B) = P(A) \cdot P(B)$ |
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**Example:** Flip a coin and roll a number cube. $P(\text{heads} \cap 6) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$

**Dependent events.** A bag has 5 red and 3 blue marbles. Draw 2 marbles.

| **With replacement** (put the first one back) | **Without replacement** (keep the first one out) |
|---|---|
| The second draw is the same as the first. Independent. | The bag changes: 4 red and 3 blue are left. Dependent. |
| $P(\text{red, red}) = \frac{5}{8} \cdot \frac{5}{8} = \frac{25}{64}$ | $P(\text{red, red}) = \frac{5}{8} \cdot \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$ |

| **General multiplication rule:** $P(A \cap B) = P(A) \cdot P(B \mid A)$<br>$P(B \mid A)$ is read "the probability of $B$ **given** $A$." It is the probability of $B$ after $A$ has already happened. |
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In the marble example: $P(\text{red second} \mid \text{red first}) = \frac{4}{7}$.

**Tip:** "At least one" questions are often easier with the complement. $P(\text{at least one wrong}) = 1 - P(\text{all right})$.

Work with a partner. Compare answers when you finish each problem.

1. These events are independent. Find each probability.

   (a) Flip a coin 3 times. $P$(3 heads)

   (b) Roll a number cube twice. $P$(two 6s)

   (c) A basketball player makes 80% of her free throws. $P$(she makes 2 in a row)

   (d) Same player. $P$(she misses both)

2. You guess on a 5-question true-or-false quiz.

   (a) $P$(all 5 right)

   (b) $P$(at least one wrong)

3. A bag has 4 green and 6 yellow marbles. Draw 2 marbles.

   (a) Find $P$(green, green) **with** replacement.

   (b) Find $P$(green, green) **without** replacement.

   (c) Why is the answer to (b) smaller?

4. Independent or dependent? Explain.

   (a) Flip a coin, then roll a number cube.

   (b) Pick a class president, then pick a vice president from the students who are left.

   (c) Draw a card, put it back, shuffle, and draw again.

   (d) It rains today, and you carry an umbrella today.

5. At a school, $P(\text{late}) = 0.2$ and $P(\text{detention} \mid \text{late}) = 0.5$.

   (a) Find $P(\text{late} \cap \text{detention})$.

   (b) Explain what your answer means in words.